

NJALA UNIVERSITY

DEPARTMENT OF BSC NURSING



BO CAMPUS

HISTOPATHOPHYSIOLOGY NOTE

ALTERATIONS IN BODY DEFENSE

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Body Defense Mechanisms against Injury

To protect against injury and infection, the body has various defense mechanism. These defense mechanisms are:-

- 1) The skin and mucous membranes
- 2) The mononuclear phagocyte system
- 3) The inflammatory response and
- 4) The immune system

1. The skin and mucous membranes

The skin and mucous membranes are the first line of defense mechanisms. They serve as a mechanical barrier for protection of the body against different injurious agents. Break

in these barriers give pave for way of injurious agents.

2. Mononuclear Phagocyte System

The mononuclear phagocyte system (MPS) consists of monocytes and macrophages and their precursor cells. In the past, the MPS system was called the reticuloendothelial system (RES). However, it is not a body system with distinctly defined tissues and organs. Rather, it consists of phagocytic cells located in various tissues and organs. The phagocytic cells are either fixed or free (mobile). The macrophages of the liver, spleen, bone marrow, lungs, lymph nodes, and nervous system (microglial cells) are fixed phagocytes. The monocytes (in blood) and the macrophages found in connective tissue, termed histolytic, are mobile, or wandering, phagocytes.

Monocytes and macrophages originate in the bone marrow. Monocytes spend a few days in the blood and then enter tissues and change into macrophages. Tissue macrophages are larger and more phagocytic than monocytes. The functions of the macrophage system include recognition and phagocytosis of foreign material such as microorganisms, removal of old or damaged cells from circulation, and participation in the immune system.

Location and Name of Macrophages

LOCATION	NAME
Connective Tissue	Histiocytes
Liver	Kupffer Cells
Lung	Alveolar Macrophages
Spleen	Free and Fixed macrophages
Bone Marrow	Fixed Macrophages
Lymph Nodes	Free and Fixed macrophages
Bone Tissue	Osteoclasts
Central Nervous System	Microglial Cells
Peritoneal Cavity	Peritoneal Macrophages
Pleural Cavity	Pleural Macrophages
Skin	Histiocyte, Langerhans' Cells
Synovium	Type A Cells

3. INFLAMMATORY RESPONSE

The inflammatory response is a sequential reaction to cell injury. It neutralizes and dilutes the inflammatory agent, removes necrotic materials, and establishes an environment suitable for healing and repair. The term *inflammation* is often but incorrectly used as a synonym for the term *infection*. Inflammation is always present with infection, but infection is

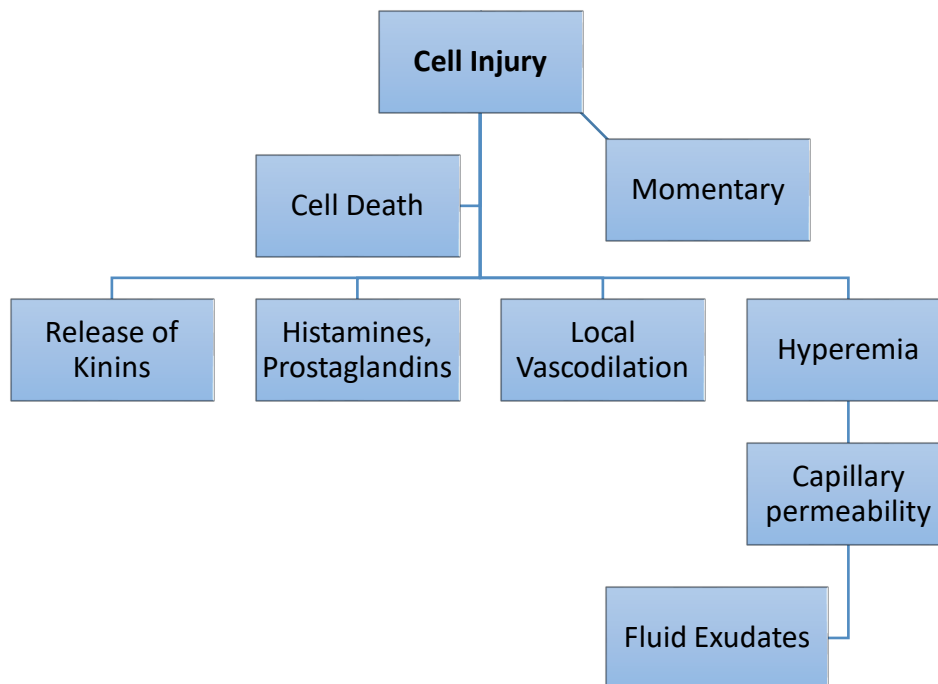
not always present with inflammation. However, a person who is neutropenic may not be able to mount an inflammatory response. An infection involves invasion of tissues or cells by microorganisms such as bacteria, fungi, and viruses. In contrast, inflammation can also be caused by nonliving agents such as heat, radiation, trauma, and allergens. The mechanism of inflammation is basically the same regardless of the injuring agent. The intensity of the response depends on the extent and severity of injury and on the reactive capacity of the injured person. The inflammatory response can be divided in to:-

- 1) Vascular response,**
- 2) Cellular response,**
- 3) Formation of exudates**
- 4) Healing.**

1) VASCULAR RESPONSE

After cell injury, arterioles in the area briefly undergo transient vasoconstriction. After release of histamine and other chemicals by the injured cells, the vessels dilate. This vasodilatation results in hyperemia (increased blood flow in the area), which raise filtration pressure. Vasodilatation and chemical mediators cause endothelial cell retraction, which increases capillary permeability. Movement of fluid from capillaries into tissue spaces is thus facilitated. Initially composed of serous fluid, this inflammatory exudates later contains plasma proteins, Primarily albumin. The proteins exert oncotic pressure that further draws fluid from blood vessels. The tissue becomes edematous.

Vascular Response in Inflammation



2) THE CELLULAR RESPONSE

This is characterized by extravasations of leucocytes from the lumen into interstitial tissue followed by phagocytosis.

Extravasations involve the following sequence of events: -

- a) Margination
- b) Transmigration across the endothelium to interstitial tissue (also called diapedesis).
- c) Migration in the interstitial tissues towards a chemotactic stimulus called Chemotaxis.

a) Margination of leukocytes

□ It is the adherence of leukocytes to the endothelial cells lining. Mainly to the post Capillary venules.

b) Transmigration of leukocytes

□ This is the movement of leukocytes by extending pseudopodia through the vascular wall by a process called **diapedesis**.

□ Leukocytes escape from venules and small veins but only occasionally from capillaries.

c) Chemotaxis

□ It is a unidirectional leukocyte attraction within tissue space guided by the presence of bacteria and cellular debris.

□ All granulocytes, monocytes and to a lesser extent lymphocytes respond to chemotactic stimuli.

d) Phagocytosis

- Once the cell has reached to the site of injurious agent (in interstitial tissue) phagocytosis ensues.
- Phagocytic cells include polymorphonuclear leukocytes (particularly neutrophils), monocytes and tissue macrophages.
- Phagocytosis involves three distinct but interrelated steps:-

1. Recognition and attachment ; of the particle to be ingested by the leukocytes:

- Phagocytosis is enhanced if the material to be phagocytosed is coated with certain plasma proteins called opsonins.

2. Engulfment

- As a result of fusion between the phagosome and lysosome, a phagolysosome is formed and the engulfed particle is exposed to the degradative lysosomal enzymes

3. Killing or degradation

- The ultimate step in phagocytosis of bacteria (any foreign body) is killing and degradation.

There are two forms of bacterial killing methods:

a. Oxygen independent mechanism:

- It is mediated by lysosomal enzymes (the primary and secondary granules) of polymorphonuclear leukocytes.

b. Oxygen dependent killing of microbes:

- This is microbial killing by the generation of oxygen metabolites such as super oxide, H₂O₂, HO⁻, etc.

Chemical mediators of inflammation

- Chemical mediators originate either from the plasma or from cells (neutrophils, macrophages, lymphocytes, basophils, mast cells and platelets). Some of the chemical mediators of inflammation include histamine, serotonin, lysosomal enzymes, prostaglandins, leukotrienes, activated oxygen species, nitric oxide, cytokines,

Mediators of Inflammation

MEDIATION	SOURCE	MECHANISM OF ACTION
Histamine	Stored in granules of basophils, mast cells, platelets	Causes vasodilation & increased vascular permeability by stimulating contraction of endothelial cells and creating widened gap between cells Causes increased vasodilation and increased vascular permeability by
Serotonin	Stored in platelets, mast cells, enterochromaffin cells of GI tract	
Kinins	Produced from precursor factor kininogen as a	

(bradykinin)	result of activation of Hageman factors (XII) of clotting system	stimulating endothelial class and creating widened gap between cells; stimulates smooth muscle contraction
Complement Components (C3a, C4a, C5a)	Anaphylaxis agents generated from complement pathway activation	Causes contraction of smooth muscle and dilation
Fibrinopeptides	Produced from activation of the clotting system	of blood vessels; result in stimulation of pain
Prostaglandins and	Produced from Arachidonic acid	Stimulate histamine release; stimulate chemotaxis
		Increases vascular permeability; stimulate chemotaxis from neutrophils and monocytes PGE1 AND PGE2 cause vasodilation; LTB4 stimulate chemotaxis

3) EXUDATES FORMATION

Exudates consist of fluid and leukocytes that move from the circulation to the site of injury. The nature and quantity of exudates depend on the type and severity of the injury and the tissues involved.

TYPES OF INFLAMMATORY EXUDATE

TYPE	DESCRIPTIN	EXAMPLES
Serous	Serous exudates results from outpouring of fluid that has low cell and protein content; it is seen in early stages of inflammation or when injury is mild	Skin blisters, pleural Effusion
Catarrhal	Catarrhal exudates are found in tissues where cells produce mucus. Mucus production is accelerated by inflammatory response.	Runny nose associated with URTI
Fibrinous	Fibrinous exudates occur	Furuncle(boil), abscess cell ulites (diffuse in

Hemorrhagic	with increasing vascular permeability and fibrinogen leakage into tissue spaces. Excessive amount of fibrin coating tissue surfaces may cause them to adhere. Hemorrhagic exudates results from rupture or necrosis of blood vessels walls; it consists of RBCs that escape into tissue.	flammation in connective tissue) Hematoma
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Clinical Manifestations of inflammations

The clinical manifestations of inflammation can be classified as;

- i. Local response to inflammation includes the manifestations of redness, heat, pain, swelling, and loss of function (see table 2.4).
- ii. Systemic response to inflammations

LOCAL MANIFESTATIONS OF INFLAMMATION

MANIFESTATIONS	CAUSE
1) Redness (rubor)	Hyperemia from vasodilatation
2) Heat (color)	Increased metabolism at inflammatory site
3) Pain (dolor)	Change in PH; Change in ionic concentration; nerve stimulation
4) Swelling (tumor)	by chemicals (e.g. histamine, prostaglandins); pressure form
5) Loss of function (function laesa)	fluid exudates Fluid shift to interstitial spaces; fluid exudates accumulation Swelling and pain

Systemic manifestations of inflammation:-

Include leukocytosis with a shift to the left, malaise, nausea and anorexia, Increased pulse and respiratory rate, and fever.

Leukocytosis results from the increased release of leukocytes from the bone marrow. An increase in the circulating number of one or more types of leukocytes

may be found. Inflammatory responses are accompanied by the vaguely defined constitutional symptoms of malaise, nausea, anorexia, and fatigue. The causes of these systemic changes are poorly understood but are probably due to complement activation and the release of cytokines (soluble factors secreted by WBCs that act as intercellular messengers) from stimulated WBCs.

Three of these cytokines, interleukin-1 (IL-1), interleukin-6 (IL-6), and tumor necrosis factor (TNF), are important in causing the constitutional manifestations of inflammation, as well as inducing the production of fever. An increase in pulse and respiration follows the rise in metabolism as a result of an increase in body temperature.

☐ **Fever**

The onset of fever is triggered by the release of cytokines. The most potent of these cytokines are IL-1, IL-6, and TNF (released from mononuclear phagocytic cells).

These cytokines cause fever by their ability to initiate metabolic changes in the temperature regulating center. The synthesis of prostaglandin E₂ (PGE₂) is the most critical metabolic change. PGE₂ acts directly to increase the thermostatic set point. The hypothalamus then activates the sympathetic branch of the autonomic nervous system to stimulate increased muscle tone and shivering and decreased perspiration and blood flow to the periphery. Epinephrine released from the adrenal medulla increases the metabolic rate.

The net result is fever. With the physiologic thermostat fixed at a higher than – normal temperature, the rate of heat production is increased until the body temperature reaches the new set point. As the set point is raised, the hypothalamus signals and

increases in heat production and conservation to raise the body temperature to the new level. At

this point the individual feels chilled and shivers. The shivering response is the body's method of

raising the body's temperature until the new set point is attained. This seeming paradox is

dramatic: the body is hot yet an individual piles on blankets and may go to bed to go warm.

When the circulating body temperature reaches the set point of the core body temperature, the

chills and warmth-seeking behavior cease.

The classifications of febrile response

☐ The febrile response is classified into four stages: Prodromal, chill, flush and defervescence.

STAGES OF THE FEBRILE RESPONSES

STAGE	CHARACTERISTICS
• Prodromal	• Nonspecific complaints such as mild headache, fatigue, general malaise, and muscle aches
• Chill	• Cutaneous

• Flush • Defervescence	vasoconstriction, “goose pimples,” pale skin; feeling of being cold; generalized shaking chill; shivering causing body to reach new temperature set by control center in hypothalamus • Sensation of warmth throughout body; cutaneous vasodilatation; warming and flushing of the skin • Sweating ; decrease in body temperature
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The released cytokines and the fever they trigger activate the body's defense mechanisms. Beneficial aspects of fever include increased killing of microorganisms, increased phagocytes by neutrophils, and increased proliferation of T cells. Higher body temperature may also enhance the activity of interferon, body's natural virus-fighting substance.

Types of Inflammation

The basic types of inflammation are acute, sub-acute, and chronic.

In acute inflammation the healing occurs in 3 to 3 weeks and usually leaves no residual damage. Neutrophils are the predominate cell type at the site of inflammation.

A sub acute inflammation has the features of the acute process but lasts longer. For example, infective endocarditis is a smoldering infection with acute inflammation, but it persists throughout weeks or months.

Chronic Inflammation lasts for weeks, months, or even years. The injurious agent persists or repeatedly injures tissue. The predominate cell types at the site of inflammation are lymphocytes and macrophages. Chronic inflammatory process are debilitating and can be devastating. The prolongation and chronicity of any inflammation may be the result of an alteration in the immune response.

4) HEALING PROCESS

The final phase of the inflammatory response is healing. Healing includes the two major components of regeneration and repair. **Regeneration** is the replacement of lost cells and

tissues with cells of the same type. **Repair** is healing as a result of lost cells being replaced by connective tissue of different origin. Repair is the more common type of healing

and usually results in scar formation. Example of healing processes:-

A) Wound Healing

B) Fracture Healing

A) Wound Healing

Wound healing follows also either Regeneration or Repair by scar.

□Regeneration

The ability of cells to regenerate depends on the cell types (see Table). Labile cells, such as epithelial cells of the skin, lymphoid organs, bone marrow, and mucous membranes of the GI, urinary, and reproductive tracts, divide constantly. Injury to these organs is followed by rapid regeneration.

Stable cells retain their ability to regenerate but do so only if the organism injured. Examples of stable cells are liver, pancreas, kidney, and bone cells.

Permanent cells do not regenerate: Examples of these cells are neurons of the CNS and cardiac muscle cells. Damage to CNS neurons or heart muscle can lead to permanent loss.

Healing will occur by repair with scar tissue.

REGENERATION ABILITY OF DIFFERENT TYPES OF TISSUES

TISSUE TYPE	REGENERATIVE ABILITY
• Epithelial Skin, linings of blood vessels, mucous membranes	• Cells readily divide and Regenerate
• Connective Tissue Bone	• Active tissue heals rapidly
Cartilage	• Regeneration possible but slow
Tendons & Ligaments	• Regeneration possible but slow
Blood	• Cells actively regenerate
• Muscle	• Regeneration usually possible (particularly in GI tract)
Smooth	• Damaged muscle replaced by connective tissue
Cardiac	
Skeletal	• Connective tissue Replaces severely

	damaged muscle; some regeneration in moderately damaged muscle occurs
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Repair is a more complex process than regeneration. Most injuries heal by connective tissue repair. Repair healing occurs by primary, secondary, or tertiary intention.

Primary Intention

Primary intention healing takes place when wound margins are nearly approximated, such as in a surgical incision or paper cut. A continuum of processes is associated with primary healing (see Table 2.7). These processes include three phases.

Phases in Primary Intention Healing

PHASE	ACTIVITY
• Initial (3 to 5 days) • Granulation (5 days to 4 weeks) • Maturation Phase & Scar Contraction (7 days to several months)	Approximation of incision edges; migration of epithelial cells ; clot serving as meshwork for starting capillary growth Migration of fibroblasts; secretion of collagen ; abundance of capillary buds; fragility of wound Remodeling of collagen; strengthening of scar

Secondary Intention

Wounds that occur from trauma, ulceration, and infection and having large amounts of exudates and wide irregular wound margins with extensive tissue loss may not have edges that can be approximated. The inflammatory reaction may be greater than in primary healing. This results in more debris, and exudates. The debris may have to be cleaned away (debrided) before healing can take place. In some instances a primary lesion may become infected, creating additional inflammation. The wound may reopen, and healing by secondary intention takes place. The process of healing by secondary intention is essential as primary healing. The major differences are the greater defect and the gaping wound edges. Healing and granulation takes place from the edges inward and from the bottom of the wound upward until the defect is filled. There is more granulation tissue and the result is much larger scar.

Tertiary intention

Tertiary intention (delayed primary intention) healing occurs with delayed suturing of a wound in which two layers of granulation tissue are sutured together. This occurs when a contaminated wound is left open and sutured closed after the infection is controlled. It also occurs when a primary wound becomes infected, is opened, is

allowed to graduate, and is then sutured. Tertiary intention usually results in a larger and deeper scar than primary or secondary intention.

B. Bone Healings

□ Bone healing occurs in a similar manner with soft tissue healing, but it is however more complex and takes longer time. Although the exact mechanism is controversial; the following five stages are identified:-

1) Stage of Hematoma Formation:-

- Occurs during the first 48-72 hours following fracture.
- Develops as blood leaks from torn vessels into the bone fragments and clotting factors in the injured area remain to initiate the formation of fibrin mesh work, which serves as frame work for in growth of fibroblast and new capillaries. Granulation tissue eventually invade and replace the clots.

2) Stage of Cellular Proliferation:-

- Three layers of bone (periosteum, endosteum, and medullary canal) are involved in the proliferation. During this process the osteoblast multiply and differentiate into fibrocartilagenous callus; softer and more flexible than callus. The collar edge on either side of the fracture eventually unit to form a bridge, which connects the bone, fragments.

3) Stage of Callus Formation:-

- During the early stage of callus formation, the fracture become sticky as osteoblast continue to move in and through the fibrin bridge to help keep it firm. Then bone calcifies as mineral salt deposit to form true callus in 3-4 weeks time.

4) Stage of Ossification:-

- It involves the final lying down of bone. This is the stage at which fracture has been bridged and is firmly united. Mature bone replaces the callus and excess callus is gradually resorbed and appear united on X-ray.
- Cast can be removed safely at this stage.

5) Stage of Remodeling and Consolidation: -

- It involves resorption of the excess bony callus that develops within the marrow space and encircle the extended aspect of fracture site.

N.B: - Function usually returns after six months after union, and complete function may take longer time.

Factors Affecting Fracture Healings

Factors that determine degree of fracture healing are classified as local factors and those factors specific to the patients: -

1) Local Factors: -

- Nature of injury
- o Edematous.

- o Displacement.
- Degree of bridge formation in the healing.
- Amount of bone loss
- Type of bone injury
 - o Cancellous bone heals faster
 - o Cortical bone heals slowly.
- Degree of immobilization achieved.
- Local infection
 - o Retard healing.
- Local malignancy
- Bone necrosis
 - o Prevents blood flow to the site.

2) Individual Factors: -

- Age:-younger –healing is faster.
- Older –healing is slower.
- Poor nutrition
- Debilitating diseases: – Diabetes
- Rheumatoid Arthritis
- Coagulation disorders.
- Circulatory problems.

4. IMMUNOLOGICAL DEFENSE MECHANISM

Introduction

Immunopathology is the study of diseases, which have or appear to have an immunologic basis. The immune system is a two-edged sword on one hand, **immunodeficiency states** render humans easy prey to infections and possibly tumors; on the other hand, **a hyperactive immune system** may cause fatal disease.

A) Hypersensitivity Reactions

Definition

- ☐ Hypersensitivity reactions refer to exaggerated response of immune system to an antigen (foreign body). The antigen that elicits the response is called an allergen.
- ☐ The purpose of the immune response is to protect against invasion by foreign organisms, but they often lead to tissue damage. Thus, an immune response that results in tissue injury is broadly referred to as **a hypersensitivity reaction**.

Classification: -

☐ ☐ Hypersensitivity reactions can be divided into 4 types depending on the mechanism of immune recognition involved and on the inflammatory mediator system recited, which is **Gell and Combs classification**.

Types - I, II, and III depend upon the interaction of antibody with the target, whereas, in type IV reactions recognition is achieved by antigen receptors on T-cells.

1) Type-I (Immediate Hypersensitivity Reaction)

□□ It affects those people who are susceptible to the disease genetically. Susceptibility to allergy is inherited and may result from excessive IgE-production. The immune response is activated when antigen binds to IgE antibodies attached to the surface of mast cells. Mast cells are found in connective tissue, skin, and mucus membranes. The reaction proceeds when the IgE molecule specific for a particular antigen become cross-linked on the surface of mast cell and triggers the release of intracellular granules. The granules contain histamine and other chemotactic substances. Histamine causes peripheral vasodilatation and an increased vascular permeability, resulting in local vascular congestion and edema. Histamine also cause bronchospasm often associated with allergic reaction.

Example: - Bronchial asthma, Allergic rhinitis, contact dermatitis

2) Type-II (Cytotoxic Hypersensitivity)

□□ In type-II hypersensitivity response, a circulating antibody, usually an IgG reacts with an antigen on the surface of a cell. The antigen could be a normal component of the membrane. It also may be a foreign antigen, such as a pharmacological agent, that adheres to the surface of the host's own cells.

□ **Example:** - Autoimmune hemolytic anemia , Erythroblastosis fetalis, Drug- induced hemolysis

3) Type-III (Immune complex disease)

□ Immune complex disease results in the formation of antigen-antibody complexes that activate a variety of serum factors especially complement. This results in precipitation of complexes in vulnerable areas, leading to inflammatory process in intravascular, synovial, endocardial and other membranes of the vulnerable organs. The antigen-antibody complexes accumulate in the circulation and if not cleared, the complex lodge in the tissue, where they initiate inflammatory reaction that leads to tissue destruction.

4. Type-IV (Cell- Mediated Hypersensitivity)

□ Type-IV response is the result of specifically sensitized T-lymphocytes. □ Activation causes a delayed-type response.

Delayed Hypersensitivity

- It is due to the specific interaction of T-cells with antigen. The T-cells reacts with the antigen and release lymphokines that draws macrophages in to the area and elicit inflammatory reaction locally. The tuberculin response is the best example of the delayed hypersensitivity response and is used to determine whether a person has been sensitized to the disease. Reddening and indurations of the site begins with in 12-72hours

- Example

Contact Dermatitis

- A common allergic skin reaction
- It is a T-cell response with a delayed reaction
- It occurs on contact with certain common household chemicals, cosmetics and plant toxins.

- The area becomes red and indurated, and vesicles begin to appear. The lymphocytes and macrophages infiltrate the area and react against the epidermal cells.

☐ **Transplant or Graft Rejection**

- Rejection of tissues and transplanted organs involves several of the hypersensitivity response. Rejection is defined as the process by which the immune system of the host recognizes, develops sensitivity to, and attempts to eliminate the antigenic differences of the donor organ.
- It is a complex reaction that involves both cell mediated and humoral responses. Cytotoxic T lymphocytes may either attack grafted tissues directly or secrete chemotactic cytokines that activate macrophages for tissue destruction. Humoral response involves formation of antibodies that circulate. The antibodies cause graft tissue destruction as they become more sensitized.

B) Immunodeficiency diseases

Definition:-

The term immunodeficiency covers a group of disorders in which defects result in impaired resistance to microbial infections.

Classifications

Deficiencies of specific immune response are divided into primary and secondary types.

1. Primary immunodeficiency diseases:-

Are those disorders which usually manifest in early childhood and there is a good evidence that the disease is genetically determined. Primary immunodeficiency diseases which are almost always genetically determined are further divided into:-

- ☐ ☐ Deficiencies of antibody (B - cells).
- ☐ ☐ Deficiency of cell mediated (T-cells) immunity.
- ☐ ☐ Combined T-cells and B-cells Deficiency (Severe combined immunodeficiency disease)
- ☐ ☐ Complement factor deficiency.

2) Secondary immunodeficiency diseases

- Secondary immunodeficiency is defined as deterioration in previously strong immune status.

- The causes are usually secondary to systemic conditions like:-

- ☐ ☐ Diabetic mellitus
- ☐ ☐ Renal failure
- ☐ ☐ Liver failure
- ☐ ☐ Poor nutrition
- ☐ ☐ Cancers
- ☐ ☐ Broad spectrum antibiotics
- ☐ ☐ Prolonged steroid therapy
- ☐ ☐ Human Immunodeficiency Virus (HIV/AIDS)